

NCERT Class 10 Science - Chapter 11: Electricity

11.1 Electric Current & Circuit

Electric Current $I = Q / t$. SI unit is Ampere (A). 1 Ampere = 1 Coulomb per second. Current flows from positive to negative terminal.

11.4 Ohm's Law & Resistance

Ohm's Law: Potential difference V across metallic wire is directly proportional to current I : $V = I * R$. Resistance $R = \rho * l / A$. SI unit is Ohm (Ω).

11.6.1 Resistors in Series

Equivalent resistance in series $R_s = R_1 + R_2 + R_3$. Current I remains constant through all series resistors.

11.6.2 Resistors in Parallel

Reciprocal equivalent resistance in parallel $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$. Potential difference V remains constant across each resistor. Remember to invert $1/R_p$ to find total resistance $R_p = (R_1 * R_2) / (R_1 + R_2)$.

11.7 Heating Effect & Joule's Law

Joule's Law of Heating: Heat produced $H = I^2 * R * t$. Heat is proportional to square of current, resistance, and time.